

Grapes

Hollis Vane

ONE

Concord

You will want to know, I suppose, why a man who has spent half a century in the company of wine should sit down, at seventy-one, to write a book about grapes.* People always want to know why. It is the first question at a dinner and the last thing anyone remembers of the answer, so I will keep it short. Wine has been written about enough. It has been praised, scored, auctioned, forged, and explained, and it has learned to talk about itself, as things do once they have been explained. The grape has not. The grape is wine before it went to school. I find, at my age, that I prefer things before they went to school. I include myself in this. We are all more interesting before we have been explained, and most of us are never given the chance to find out.

So: the grape. Let us begin where the encyclopedias begin, since that is where I began, and since a man who has spent his life with his nose in a glass ought to be made to look things up.

The grape belongs to the genus *Vitis*, in a family of climbing plants called the Vitaceae.¹ There are sixty to eighty species of *Vitis*,

* Card 1. Tuesday, 7 January 2020, 11:52 p.m. I begin, as I mean to go on, with a card.

¹The family is large and mostly unknown to anyone who is not a botanist: the

depending on whom you ask and on how closely they have looked at the vines of Asia.² They are native to the northern temperate zone, and most of them live in eastern Asia and in North America. Europe and western Asia have exactly one. That one is *Vitis vinifera*, the wine-bearing vine, and it supplies the great majority of the grapes grown anywhere in the world. It is the grape of every wine you have heard of. It is also, in the form of the wild vine from which it was taken, a plant that comes in two sexes. The wild vine is male or female, and the female needs a male nearby before she will set fruit.³ The cultivated vine is, for the most part, both at once. Its flowers carry everything they need. It does not require anybody. I have sometimes envied it.

How many kinds of grape are there? Ten thousand, according to the same international organisation, of which thirteen cover more

Encyclopædia Britannica counts twelve genera in it, and a group of botanists writing in 2018 (Wen and others, in the *Journal of Systematics and Evolution*) count sixteen genera and about nine hundred and fifty species. Virginia creeper is a cousin. So is Boston ivy, which covers so many old brick walls, and which I have always suspected of being there to keep us from asking what is underneath.

²Sixty to eighty is the *Britannica*'s range. The International Organisation of Vine and Wine, which is to the grapevine roughly what the Vatican is to the saints, says eighty, on the authority of a 2007 paper by Wen and others. The disagreement is mostly about the Asian species, which are many and poorly separated. I have kept a card for each number. I do not yet know which card to throw away.

³The wild form is *Vitis vinifera* subspecies *sylvestris*; the cultivated form is subspecies *vinifera*. The usual account, as I write, puts the domestication in western Asia some eight thousand years ago, and the genetic evidence also points to the Near East (Myles and others, *Proceedings of the National Academy of Sciences*, 2011). On the sexes: Vasconcelos and colleagues, in the *American Journal of Enology and Viti-culture* (2009), and Picq and colleagues, *BMC Plant Biology* (2014).^a

^aPicq and his colleagues trace the difference to a small region of a single chromosome, which I will not pretend to understand, except to say that the whole history of the vineyard comes down, as so many histories do, to a very small thing that nobody noticed for a very long time.

than a third of the world's vineyards and thirty-three cover half.⁴ Ten thousand kinds, and half the vineyards of the world planted to thirty-three of them. We are loyal, all of us, to the first few things we were given, and we call the loyalty taste.

And how much grape is there? In 2018, the last year for which I have the figures, the world grew 77.8 million metric tonnes of grapes.⁵ That is a great deal of fruit to have so little written about it that is not about something else.

I bought a bunch of grapes yesterday at the market in St. Helena. It is January, and nothing in this valley is ripe; the vines outside my window are bare wood. The box said *Product of Chile*, which is to say that it was summer where these grapes were picked, and they had come north in the dark of a ship to be eaten in the winter of strangers.

⁴International Organisation of Vine and Wine, *Distribution of the World's Grapevine Varieties* (2017). The same paper adds that the standard catalogue of the subject, the *Vitis International Variety Catalogue*, held 21,045 names, 12,250 of them for *vinifera*, but that many of these are the same grape under different names, and that the true number of *vinifera* varieties is nearer six thousand.^a

⁵International Organisation of Vine and Wine, *2019 Statistical Report on World Vitiviniculture*. Of the 77.8 million tonnes, 57 per cent went to wine (and to musts and juices), 36 per cent were eaten as table grapes, and 7 per cent were dried. The dried share is counted as fresh grapes, before drying, at about four kilograms of grapes to one kilogram of raisins. The largest growers, in millions of tonnes: China 11.7, Italy 8.6, the United States 6.9, Spain 6.9, France 6.2. The world's vineyards covered 7.4 million hectares.^b

^aA grape that travels collects names, as people do who travel. The same vine may be one thing in its own valley and another two valleys over, and a third thing in a nurseryman's catalogue in another language. The catalogue's work is to find out which names belong to one plant. It is slow work. It is also, I think, a kind of kindness.

^bIn pounds, which is how I was raised to think and how I still think when I am tired, 77.8 million tonnes is about 171.5 billion pounds. Divided among the 7.7 billion of us the United Nations counted in 2019, it comes to 10.1 kilograms, or 22.3 pounds, a head. I have done the division three times. It comes out the same each time, which is more than I can say for some of my counting (see 200, p. 4).

I paid for them, carried them home, and put them on the kitchen scale. They weighed 612 grams.* Then I counted them.

☛ There were 117 berries. I counted them again, to be sure, and there were 118. I am a careful man; I have been paid, for most of my life, to be careful; and I could not count a bunch of grapes twice and get the same number. I counted a third time, slowly, pulling each berry a little aside with the tip of a pencil, and there were 117. I have decided to believe it. It is the number I got most often, and in the end that is what most of us mean by the truth.⁶

Then, because I was not tired, I cut eleven of them in half with a paring knife on a white plate. Botanically, a grape is a true berry. It is a single ovary grown fat: the skin is the outer wall, the flesh is the middle, and the thin layer round the seeds is the inner wall, and there is no stone. The skin wears a dull, waxy bloom that rubs off on the thumb; it is there to keep the water in. The ovary has two small chambers, and each chamber holds two ovules, so that a grape can never have more than four seeds.⁷ These were sold as seedless, and the box was right. Eleven times I counted the seeds in a grape, and eleven

*The scale is a Salter, in metal, older than the house. I trust it more than I trust myself. I have had it longer.

⁶The bunch has a botanical name and an anatomy. It is a panicle: a branched stalk carrying many flowers, and later many fruits. The part that joins the bunch to the cane is the peduncle; the main axis is the rachis; and each berry hangs on its own short stem, the pedicel. (The terms are in Winkler's *General Viticulture*, the textbook I was handed when I came to California, and which I have carried around since.) A single cluster may carry as many as a thousand flowers. If one in three of them becomes a berry, the growers are content.^a

⁷Vasconcelos and others, *American Journal of Enology and Viticulture* (2009), on the flower; and the standard extension manuals, on the seeds: one to four in the seeded varieties, never five. I have looked. I have not found a fifth.

^aBy that arithmetic my 117 berries were once some 350 flowers, and some 230 of those flowers came to nothing. Nobody counted them either. We count the ones that set.

times there were none. It is restful, at the end of a day, to count to zero.*

The first grape I ever knew was not a grape. It was a jelly.

I grew up in Iowa, in a town I see no reason to name. Name it and someone from there will write to correct me about the year the elevator burned; leave it unnamed and it can be any of a thousand towns, which is nearer the truth. They were all the same town. There was a grain elevator that you could see before you could see anything else, a main street that ran one way into corn and the other way into corn, a church, a café, and a hardware store, which was my father's. In August the heat came up out of the fields and stood in the street, and the only cool place in town was the back of my father's store, among the nail kegs, where it smelled of oil and iron and binder twine. There were no vines. If you had asked me, at ten, what a grape was, I would have pointed at a jar.

The jar said *Welch's*. The jelly inside it was purple, nearly black in the jar and a bright, translucent violet when you spread it thin, which my mother did not, on white bread, with peanut butter.⁸ In the early 1960s Welch's put its jelly in drinking glasses printed with the Flintstones, and when the jelly was gone the glass was yours. By then I was too old to admit that I wanted one, and so I drank from

*It is 1:47 in the morning. I keep a farmer's hours, but from the wrong end of the night.

⁸It is generally believed that the peanut-butter-and-jelly sandwich was invented by American soldiers in the Second World War. It was not. The first published recipe appeared in 1901, in the *Boston Cooking-School Magazine*, by Julia Davis Chandler, and it called for currant or crab-apple jelly, not grape. It became the lunch of American children in the Depression, when bread came sliced and peanut butter came cheap. (Steve Estes, "PB&J," *Gastronomica*, 2017, and the food historian Andrew F. Smith.) The soldiers ate it, too; they did not invent it. Most of what we believe about where things come from is like this: true about the people, wrong about the dates.

Fred's face only when nobody was in the kitchen.*

That jelly is made from one grape, the Concord, and the taste of it is the taste that an American of my age means when he says "grape." It is not the taste of the grape of wine. A European, given a spoon of it, will tell you, politely, that it tastes of something else. The word for the something else is *foxy*. The French, who borrowed it from the English "fox grape," say *foxé*, and so, in my hearing, did everyone in the wine trade, with a small movement of the nose that I learned in my twenties to make myself. The molecule that is chiefly responsible has a name: methyl anthranilate.⁹ I made that small movement of the nose for fifty years. I am not proud of it. It was the first thing I learned to despise, and I learned to despise it in order to be admitted somewhere, which is how most contempt is learned.

The Concord has a maker, and his name was Ephraim Wales Bull.

He was born in Boston in 1806 and moved to Concord, Massachusetts, in 1836, to a house on Lexington Road. By his own account he sowed the seeds of a wild grape in the autumn of 1843, chose among the seedlings, and waited. His vine first bore its fruit in 1849. He showed the grape to the Massachusetts Horticultural Society in 1853, and in 1854 C. M. Hovey of Boston put it on the market.¹⁰ It was the right grape for the place. What the country wanted, Bull said,

*Cards tonight for the glasses, for Welch, for the sandwich, and for Bull (several). They go into a shoebox marked 2020, behind Card 1, which until tonight looked very lonely there.

⁹Methyl anthranilate was first found in grape juice in 1921, by Power and Chestnut, in the *Journal of the American Chemical Society*; they found it in the American grapes, the Concord and its kin, and not in the European. A later paper (Robinson and others, *American Journal of Enology and Viticulture*, 2014) still calls it the compound "considered to be responsible for the distinctive 'foxy' aroma and flavor" of the Concord. Other compounds help; this one leads.

¹⁰The dates are in the Concord Free Public Library's papers on Bull and in the histories kept by the United States Department of Agriculture's National Agricultural Library. The 1843 sowing is Bull's own account, quoted at second hand; he also spoke

was “a good table and wine grape, which shall also be early, hardy and prolific,” and the grapes people had been trying to grow in New England either died in the winter or failed to ripen before the first frost. The Concord ripened early and lived through the winters. Americans planted it widely.¹¹ Bull served in the state legislature. He did not grow rich.¹² He died in 1895, in the Concord Home for the Aged, and he is buried in Sleepy Hollow Cemetery, under a stone that says *He sowed, others reaped*.¹³

Others reaped. The chief of them was a dentist. Thomas Bramwell Welch was an Englishman, a Methodist, and a man who believed that alcohol was the ruin of his neighbours, and he practised dentistry in Vineland, New Jersey. In 1869 he heated the juice of Concord grapes to keep it from fermenting, so that his church could take communion without taking wine.¹⁴ His juice was the first of the products that put his family’s name into American kitchens, my mother’s among them.

We put up statues to the man who sold the thing. The man who

of raising two generations of seedlings, which does not fit easily into six years. Where the sources disagree I have given the date they agree on, and kept the others on cards. I have a great many cards about Ephraim Bull.

¹¹The Concord is usually called a variety of *Vitis labrusca*, the American fox grape. It is not quite. Its parentage includes some *vinifera*: Huber and colleagues (*Vitis*, 2016) make it an offspring of the old American variety ‘Catawba,’ which has European blood, and of an unknown wild parent.^a

¹²So it is said, by nearly everyone who writes about him. I have not seen his accounts.

¹³That is the wording. The punctuation varies from one source to the next: a full stop, a comma, nothing at all. I have not seen the stone. I mean to.

¹⁴Welch’s own history says he used forty pounds of Concord grapes from the vines in front of his house. In 1897 the business built a large juice plant in Westfield, New York, on the shore of Lake Erie, in the strip of Concord vineyards along the lake that is still called the Grape Belt. Its grape spread, called Grapelade, dates from 1918, and its grape jelly, by the account of the histories I have, from 1923. It still makes the jelly from Concord grapes. It says so on the jar.

^aSo the most American of grapes is not entirely American. Few of us are, if you go back far enough, and most of us have the good sense not to.

grew it gets a sentence on a stone, and the sentence has a comma in it that nobody can agree on. I do not say this to blame Dr. Welch. He reaped what was there to be reaped, and he reaped it for the Lord, which is more than most of us can say of our harvests. I say it because it is the way of things, and because the man who notices it usually does so, I have found, at the moment it begins to apply to himself.

It is late, and the bunch is still on the kitchen table, in its plastic box, minus eleven berries. In the morning I will count it again. I expect 106. If I get 106, I will believe it at once. If I do not, I will count until I do, and then I will believe that.*

*Card 23, 2:31 a.m. Lights out. The lock checked.